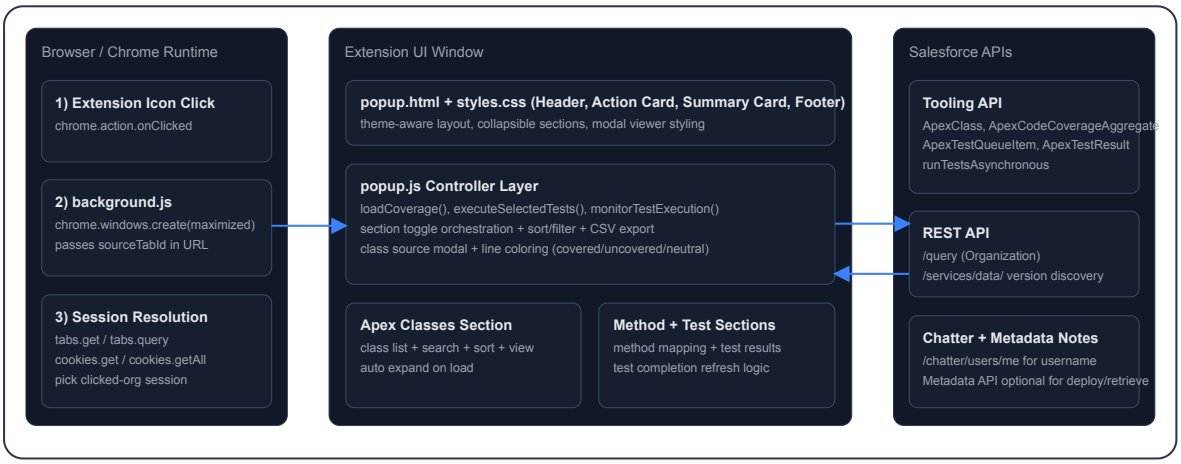
Apex Coverage Viewer - UI Architecture Diagram

Visual architecture of the extension, implemented UI logic, and API contracts with examples.

3	5	6+	5
Runtime Layers	Core UI Sections	Salesforce Endpoints	Chrome APIs

UI architecture diagram

End-to-End Flow



Implemented UI Logic

Flow	Logic Implemented	User-Visible Result
Launch and context	background passes sourceTabId; popup resolves session from clicked Salesforce tab first	Correct org session even with multiple Salesforce tabs open
Coverage load	query ApexClass + ApexCodeCoverageAggregate, then filter/sort/render	Apex Classes expands and summary metrics refresh
Method-wise mode	method mapping loaded from ApexCodeCoverage and class source parsing	Method Coverage table shows test methods per Apex method
Execute tests	runTestsAsynchronous + queue polling + failed result fetch	Live execution status with failure details
Post execution refresh	coverage reload always triggered after completion (pass or fail); section-state orchestration applied	Apex Classes and Test Execution remain expanded; Method Coverage collapses
Class viewer modal	fetch full Apex class source and color all lines by coverage status	VS Code-style dark viewer with sticky line numbers

API Inventory and Examples

Chrome Extension APIs

API	Used In	Purpose	Example
chrome.action.onClicked	background.js	Start extension from toolbar click	onClicked((tab) => ...)
chrome.windows.create	background.js	Open extension in maximized window	windows.create({ url, state: "maximized" })
chrome.tabs.get / query	popup.js	Resolve preferred Salesforce tab	tabs.get(sourceTabId), tabs.query({ active: true })
chrome.cookies.get / getAll	popup.js	Read Salesforce sid cookie for auth	cookies.get({ url, name: "sid" })
chrome.storage.local	popup.js	Persist UI preferences only	storage.local.set({ apexCoverageConfig: {...} })

Salesforce REST/Tooling/Chatter APIs

Use Case	Method + Endpoint	Primary Objects	Ex
API version discovery	GET /services/data/	Version list	GE or
Load class list	GET /services/data/v60.0/tooling/query?q=...	ApexClass	SE FR
Load class coverage	GET /services/data/v60.0/tooling/query?q=...	ApexCodeCoverageAggregate	SE Nu Nu Ap
Method mapping	GET /services/data/v60.0/tooling/query?q=...	ApexCodeCoverage	SE Te Ap Ap
Queue tests	POST /services/data/v60.0/tooling/runTestsAsynchronous	Async test run job	Bo {"
Execution status	GET /services/data/v60.0/tooling/query?q=...	ApexTestQueueItem, ApexTestResult	SE Ap Ap
Header user context	GET /services/data/v60.0/chatter/users/me	Chatter user info	GE
Header environment context	GET /services/data/v60.0/query?q=...	Organization	SE In LI

Metadata API (Optional Extension Scope)

Current implementation does not require Metadata API for runtime coverage rendering. It can be added for deployment/retrieve workflows using SOAP endpoint `/services/Soap/m/<version>`.

Operation	Endpoint	When to Use	Skeleton
retrieve()	POST /services/Soap/m/60.0	Download metadata package	<retrieve><retrieveRequest><apiVersion>60.0</apiVersion>...</retrieveRequest></retrieve>
deploy()	POST /services/Soap/m/60.0	Deploy metadata zip to org	<deploy><ZipFile>base64Zip</ZipFile><DeployOptions>...</DeployOptions></deploy>

Coverage Load | Live Test Execution | Section State Orchestration | Method Mapping